



Hewlett Packard
Enterprise

GPU - InfoROM Recovery

Objective

Sites that experience InfoROM corruption on NVIDIA GPU cards should attempt the recovery process in this article before considering replacing the GPU.

Steps

Original Cray Article Number : 000004882

This tool can be used to recover the factory backup image of the InfoROM from any board in the NVIDIA Tesla family.

NOTE: This restores the factory image such that, ECC information and valid retired pages will be lost from restoring the original factory image to the board.

The following is an example of an InfoROM corruption error message as shown from a single K40 GPU system.

```
[root@nvidia-r004 x86]# nvidia-smi
Thu Mar 10 21:43:50 2016
+-----+
| NVIDIA-SMI 340.87   Driver Version: 340.87     |
+-----+-----+
| GPU  Name      Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|      Memory-Usage | GPU-Util  Compute M. |
|=====+=====+
| 0  Tesla K40m      Off | 0000:02:00.0  Off |           ERR! | <<< (InfoROM is
Corrupted.)
| N/A  22C   P0   66W / 235W |  55MiB / 11519MiB |   100%    Default |
+-----+-----+
+-----+
| Compute processes:                       GPU Memory |
| GPU      PID  Process name                  Usage      |
+=====+=====+
| No running compute processes found          |
+-----+
WARNING: infoROM is corrupted at gpu 0000:02:00.0 [root@nvidia-r004 x86]#
```

InfoROM recovery process:

1. Files Needed [\(Put InfoROM Recovery tools \(nvflash, iromflash_ext\) in node boot image first.\)](#)

```
[root@prod-0002 x86]# pwd /global/cray/gpu_info/nvflash/x86 [root@prod-0002
x86]# ls -l total 5316 -rwxr-xr-x 1 root root 1950472 Mar 10 21:37
iromflash_ext <<< (This is the Linux X86 iromflash_ext tool) -rwxr-xr-x 1
root root 3224056 Mar 10 21:37 nvflash [root@prod-0002 x86]#
```

2. Query for NVIDIA modules.



```
[root@prod-0002 x86]# lsmod | grep nvidia nvidia_uvm 28216 0 nvidia 10692698
1 nvidia_uvm i2c_core 31084 4 nvidia,i2c_i801,igb,i2c_algo_bit
```

3. It is necessary to unload the NVIDIA module.

```
[root@prod-0002 x86]# modprobe -r nvidia_uvm or if there is no nvidia_uvm
[root@prod-0002 x86]# modprobe -r nvidia
```

4. Read the bad GPU's InfoROM:

- nvflash -b <filename of your choice>.rom
- (The following is a sample output from a Hydra/CS-Storm system that shows 8 K80 GPU cards. Each K80 GPU card is populated with 2 GK210B GPU's installed on the card. The K80 in location 8B had indicated the InfoROM was corrupted so use the following command to backup the VBIOS before flashing.)

```
[root@prod-0002 x86]# ./nvflash -b inforom_test_8B.031016.rom --entire
NVIDIA Firmware Update Utility (Version 5.265.0) Simplified Version For OEM Only
Select display adapter: <0> PLX (8747h) (10B5,8747,10B5,8747) H:--:NRM
S:00,B:02,PCI,D:00,F:00 <1> PLX (8747h) (10B5,8747,10B5,8747) H:03:SP8
S:00,B:04,PCI,D:00,F:00 <2> Tesla K80 (10DE,102D,10DE,106C) H:05:SP8
S:00,B:06,PCI,D:00,F:00 <3> Tesla K80 (10DE,102D,10DE,106C) H:05:SP16
S:00,B:07,PCI,D:00,F:00 <4> PLX (8747h) (10B5,8747,10B5,8747) H:03:SP16
S:00,B:08,PCI,D:00,F:00 <5> Tesla K80 (10DE,102D,10DE,106C) H:09:SP8
S:00,B:0A,PCI,D:00,F:00 <6> Tesla K80 (10DE,102D,10DE,106C) H:09:SP16
S:00,B:0B,PCI,D:00,F:00 <7> PLX (8747h) (10B5,8747,10B5,8747) H:--:NRM
S:00,B:0C,PCI,D:00,F:00 <8> PLX (8747h) (10B5,8747,10B5,8747) H:0D:SP8
S:00,B:0E,PCI,D:00,F:00 <9> Tesla K80 (10DE,102D,10DE,106C) H:0F:SP8
S:00,B:10,PCI,D:00,F:00 <10> Tesla K80 (10DE,102D,10DE,106C) H:0F:SP16
S:00,B:11,PCI,D:00,F:00 <11> PLX (8747h) (10B5,8747,10B5,8747) H:0D:SP16
S:00,B:12,PCI,D:00,F:00 <12> Tesla K80 (10DE,102D,10DE,106C) H:13:SP8
S:00,B:14,PCI,D:00,F:00 <13> Tesla K80 (10DE,102D,10DE,106C) H:13:SP16
S:00,B:15,PCI,D:00,F:00 <14> PLX (8747h) (10B5,8747,10B5,8747) H:--:NRM
S:00,B:82,PCI,D:00,F:00 <15> PLX (8747h) (10B5,8747,10B5,8747) H:83:SP8
S:00,B:84,PCI,D:00,F:00 <16> Tesla K80 (10DE,102D,10DE,106C) H:85:SP8
S:00,B:86,PCI,D:00,F:00 <17> Tesla K80 (10DE,102D,10DE,106C) H:85:SP16
S:00,B:87,PCI,D:00,F:00 <18> PLX (8747h) (10B5,8747,10B5,8747) H:83:SP16
S:00,B:88,PCI,D:00,F:00 <19> Tesla K80 (10DE,102D,10DE,106C) H:89:SP8
S:00,B:8A,PCI,D:00,F:00 <20> Tesla K80 (10DE,102D,10DE,106C) H:89:SP16
S:00,B:8B,PCI,D:00,F:00 <<< (Read this Bad GPU's InfoROM.) <21> PLX
(8747h) (10B5,8747,10B5,8747) H:--:NRM S:00,B:8C,PCI,D:00,F:00 <22> PLX
(8747h) (10B5,8747,10B5,8747) H:8D:SP8 S:00,B:8E,PCI,D:00,F:00 <23> Tesla
K80 (10DE,102D,10DE,106C) H:8F:SP8 S:00,B:90,PCI,D:00,F:00 <24> Tesla
K80 (10DE,102D,10DE,106C) H:8F:SP16 S:00,B:91,PCI,D:00,F:00 <25> PLX
(8747h) (10B5,8747,10B5,8747) H:8D:SP16 S:00,B:92,PCI,D:00,F:00 <26>
Tesla K80 (10DE,102D,10DE,106C) H:93:SP8 S:00,B:94,PCI,D:00,F:00 <27>
Tesla K80 (10DE,102D,10DE,106C) H:93:SP16 S:00,B:95,PCI,D:00,F:00 Select
a number (ESC to quit): 20 Adapter: Tesla K80 (10DE,102D,10DE,106C)
H:89:SP16 S:00,B:8B,PCI,D:00,F:00 The display may go *BLANK* on and off for up
to 10 seconds during access to the EEPROM depending on your display adapter and
output device. Identifying EEPROM... EEPROM ID (EF,3013) : WBond W25X40A 2.7-
3.6V 4096Kx1S, page Reading adapter firmware image... Saving of image
completed. [root@prod-0002 x86]# [root@prod-0002 x86]# ls -ltr total 5188 -rwxr-
xr-x 1 root root 1950472 Mar 10 21:37 iromflsh_ext -rwxr-xr-x 1 root root 3224056
Mar 10 21:37 nvflash -rw-r--r-- 1 root root 62348 Mar 10 22:45 nvidia-bug-
report.log.gz -rw-r--r-- 1 root root 524288 Mar 10 23:15
inforom_test_8B.031016.rom [root@prod-0002 x86]#
```

5. Run iromflsh_ext and select a GPU with a corrupted InfoROM to recover it.

```
[root@prod-0002 x86]# ./iromflash_ext -R NVIDIA InfoROM Flash Utility (External)
Version 3.45 Copyright (C) 2015, NVIDIA Corporation. All Rights Reserved. Select
display adapter: <0> PLX (8747h) (10B5,8747,10B5,8747) H:--:NRM
S:00,B:02,PCI,D:00,F:00 <1> PLX (8747h) (10B5,8747,10B5,8747) H:03:SP8
S:00,B:04,PCI,D:00,F:00 <2> Tesla K80 (10DE,102D,10DE,106C) H:05:SP8
S:00,B:06,PCI,D:00,F:00 <3> Tesla K80 (10DE,102D,10DE,106C) H:05:SP16
S:00,B:07,PCI,D:00,F:00 <4> PLX (8747h) (10B5,8747,10B5,8747) H:03:SP16
S:00,B:08,PCI,D:00,F:00 <5> Tesla K80 (10DE,102D,10DE,106C) H:09:SP8
S:00,B:0A,PCI,D:00,F:00 <6> Tesla K80 (10DE,102D,10DE,106C) H:09:SP16
S:00,B:0B,PCI,D:00,F:00 <7> PLX (8747h) (10B5,8747,10B5,8747) H:--:NRM
S:00,B:0C,PCI,D:00,F:00 <8> PLX (8747h) (10B5,8747,10B5,8747) H:0D:SP8
S:00,B:0E,PCI,D:00,F:00 <9> Tesla K80 (10DE,102D,10DE,106C) H:0F:SP8
S:00,B:10,PCI,D:00,F:00 <10> Tesla K80 (10DE,102D,10DE,106C) H:0F:SP16
S:00,B:11,PCI,D:00,F:00 <11> PLX (8747h) (10B5,8747,10B5,8747) H:0D:SP16
S:00,B:12,PCI,D:00,F:00 <12> Tesla K80 (10DE,102D,10DE,106C) H:13:SP8
S:00,B:14,PCI,D:00,F:00 <13> Tesla K80 (10DE,102D,10DE,106C) H:13:SP16
S:00,B:15,PCI,D:00,F:00 <14> PLX (8747h) (10B5,8747,10B5,8747) H:--:NRM
S:00,B:82,PCI,D:00,F:00 <15> PLX (8747h) (10B5,8747,10B5,8747) H:83:SP8
S:00,B:84,PCI,D:00,F:00 <16> Tesla K80 (10DE,102D,10DE,106C) H:85:SP8
S:00,B:86,PCI,D:00,F:00 <17> Tesla K80 (10DE,102D,10DE,106C) H:85:SP16
S:00,B:87,PCI,D:00,F:00 <18> PLX (8747h) (10B5,8747,10B5,8747) H:83:SP16
S:00,B:88,PCI,D:00,F:00 <19> Tesla K80 (10DE,102D,10DE,106C) H:89:SP8
S:00,B:8A,PCI,D:00,F:00 <20> Tesla K80 (10DE,102D,10DE,106C) H:89:SP16
S:00,B:8B,PCI,D:00,F:00 <<< (This is the GPU with corrupt InfoROM.) <21> PLX
(8747h) (10B5,8747,10B5,8747) H:--:NRM S:00,B:8C,PCI,D:00,F:00 <22> PLX
(8747h) (10B5,8747,10B5,8747) H:8D:SP8 S:00,B:8E,PCI,D:00,F:00 <23> Tesla K80
(10DE,102D,10DE,106C) H:8F:SP8 S:00,B:90,PCI,D:00,F:00 <24> Tesla K80
(10DE,102D,10DE,106C) H:8F:SP16 S:00,B:91,PCI,D:00,F:00 <25> PLX (8747h)
(10B5,8747,10B5,8747) H:8D:SP16 S:00,B:92,PCI,D:00,F:00 <26> Tesla K80
(10DE,102D,10DE,106C) H:93:SP8 S:00,B:94,PCI,D:00,F:00 <27> Tesla K80
(10DE,102D,10DE,106C) H:93:SP16 S:00,B:95,PCI,D:00,F:00 Select a number (ESC to
quit): 20 Tesla K80 (10DE,102D,10DE,106C) H:89:SP16 S:00,B:8B,PCI,D:00,F:00
Identifying EEPROM... EEPROM ID (EF,3013) : WBond W25X40A 2.7-3.6V 4096Kx1S,
page The existing InfoROM image is valid. Restoring the InfoROM from the embedded
recovery image may cause some data loss. ARE YOU SURE YOU WANT TO
CONTINUE? PRESS 'y' TO CONTINUE OR ANY OTHER KEY TO REJECT : y The display
may go *BLANK* on and off for up to 10 seconds or more during the update process
depending on your display adapter and output device. Identifying EEPROM...
EEPROM ID (EF,3013) : WBond W25X40A 2.7-3.6V 4096Kx1S, page NOTE: Preserving
straps from original image. Clearing original firmware image... . Storing updated
firmware image... ..... Verifying update... Update successful. Firmware
image has been updated from version 80.21.1B.00.02 to 80.21.1B.00.02. A reboot
is required for the update to take effect. <<<
(RESET GPU or REBOOT THE NODE!!.) InfoROM image recovered successfully.
```

The recovery files [iromflash_ext](#) and [NVFlash](#) are attached to this article below under "Attachments" if needed.

Attachments



[NVFlash](#)

[iromflash_ext](#)

